

Neel Suyog Harip

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EDUCATION

University of Illinois Urbana-Champaign

Master of Computer Science

Aug 2024 – Dec 2025

Illinois, USA

Relevant Coursework: Database Systems, Applied ML, Advanced Data Management, Text Information Systems, Cloud Computing Applications, Systems for GenAI, ML for Software Engineering, Web Programming

TECHNICAL SKILLS

Languages: Python, Java, C#, SQL, JavaScript, C++, TypeScript

Backend: Node.js, FastAPI, REST APIs, Flask

Databases & Frontend: PostgreSQL, MySQL, MongoDB, React.js, Angular

Cloud & DevOps: AWS (Lambda, S3, DynamoDB, Kinesis, API Gateway), GCP, Docker, Jenkins, CI/CD

Applied AI: LLM Integration, RAG, Prompt Engineering, LangChain, Hugging Face Transformers, Vector Embeddings, Semantic Search, OpenAI APIs, Claude Code

EXPERIENCE

Full Stack Developer

University of Illinois Urbana Champaign | Tech Stack - React.js, Node.js, PostgreSQL

Jan 2025 – May 2025

Illinois, USA

- Built a 0→1 full-stack research platform, designing and implementing the system from scratch to manage and explore 300+ research projects, improving accessibility and centralized discovery of research data
- Built and deployed semantic search using Hugging Face Transformers and vector embeddings across 300+ research projects, significantly improving discovery across structured and unstructured research data

Software Development Engineer

Tata Consultancy Services | Tech Stack: C#/.NET, SQL, JavaScript

Aug 2021 – Jul 2024

Pune, India

- Developed backend services and REST APIs for a knowledge platform supporting 500+ users, improving information retrieval speed by 30% and enabling faster operational workflows
- Saved 1,500+ hours annually and reduced manual effort by 80% by building custom C#/.NET modules and API-driven workflows to automate business processes
- Designed and maintained CI/CD pipelines using Git and Jenkins, incorporating automated validation to improve deployment reliability and reduce release errors across environments
- Implemented monitoring and alerting solutions using Splunk, improving system observability and reducing incident resolution time by 40% in production environments

Full Stack Intern

Wisdom Sprouts | Tech Stack - Angular, Node.js, Express.js, MongoDB, PHP

Nov 2019 – Jun 2020

Pune, India

- Improved application performance by reducing initial page load time by 15% using Angular lazy loading, and reduced API response size by 20% through optimized Node.js services and MongoDB queries

PROJECTS

AI Ops Automation System

Python, FastAPI, OpenAI API, Docker

Mar 2025 – Apr 2025

- Built an AI-driven automation platform that ingests logs/events, performs LLM-based analysis, and generates actionable insights using prompt engineering and structured output parsing for real-time operational workflows
- Designed backend pipelines integrating APIs, event streams, and orchestration logic to automate decision-making, with containerized deployment using Docker for scalable and reliable execution

Real-Time Financial Data Pipeline

Python, Apache Spark, Kafka, InfluxDB, Grafana

Nov 2025 – Dec 2025

- Developed a high-throughput data pipeline processing 10,000+ events/sec using Kafka and Spark for real-time ingestion and transformation of streaming data
- Designed scalable streaming architecture with time-series storage and observability using InfluxDB and Grafana, ensuring system reliability, performance monitoring, and fault detection

Cloud-Based Real-Time Data Processing Platform

AWS (Kinesis, Lambda, API Gateway, DynamoDB, S3, CloudWatch)

Feb 2025 – May 2025

- Implemented a serverless, event-driven data processing platform to ingest and transform high-volume streaming data, supporting scalable analytics and low-latency access
- Designed a serverless, event-driven architecture integrating S3, DynamoDB, and API Gateway to support historical querying, low-latency data access, and automated monitoring of streaming workloads